X25519 fixed-radix correctness validation

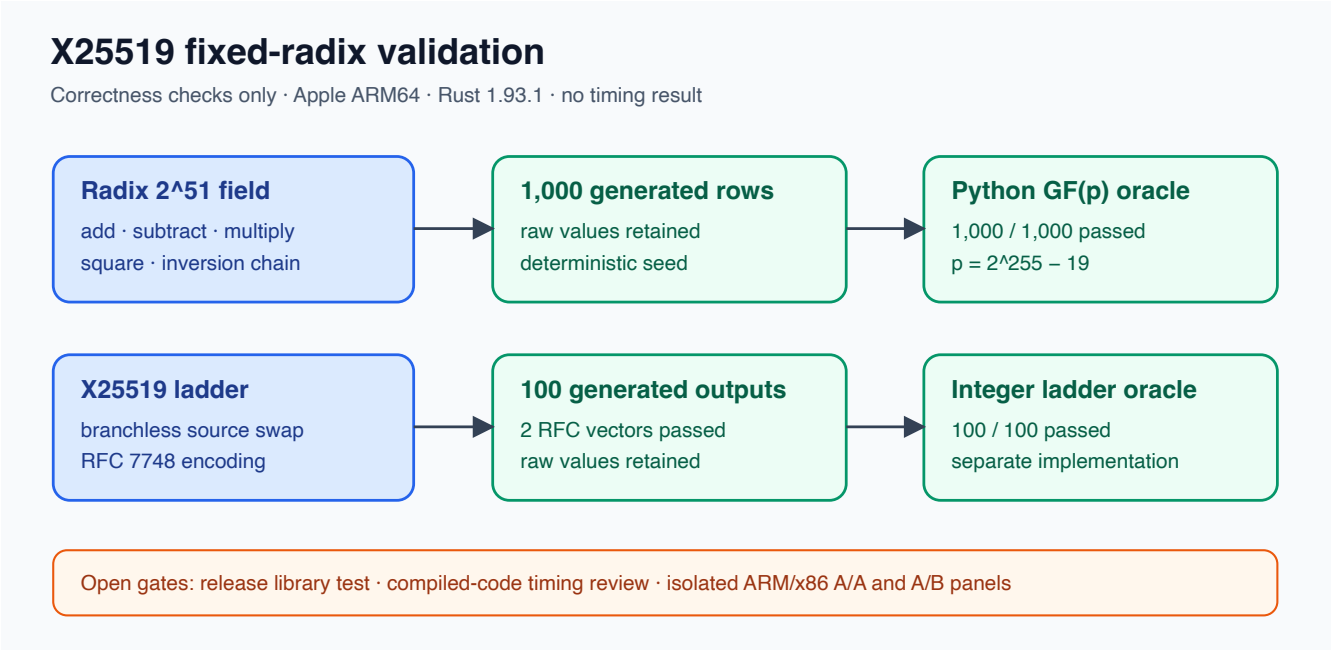
Status: observations from a local correctness run. No performance or constant-time conclusion is drawn here. The exact commands, source hashes, drivers and raw rows are in `research/ecc_arithmetic_20261013/x25519_validation/README.md`.

Implementation boundary

The public X25519 entry point now uses five radix-2^51 limbs, a dedicated square, a fixed inversion chain, and the RFC 7748 ladder with source-level branchless swaps. The original `BigUint` field remains a separate reference path; Ed25519 is unchanged. The source was uncommitted relative to repository HEAD `9f588c4f4` during this run.

Observations

On Apple ARM64 T6041 with Rust 1.93.1 and Python 3.13.1, the actual Rust field source emitted 1,000 deterministic operand-pair rows. The separate Python integer checker matched addition, subtraction, multiplication and inversion modulo `2^255 - 19` in all rows. The actual Rust X25519 source passed two RFC 7748 section 5.2 vectors, then emitted 100 deterministic full-operation outputs. A separately written Python integer ladder matched all 100. Inputs and outputs are retained as `field_rows.txt` and `x25519_rows.txt`.



An ARM64 assembly scan found two field-limb-dependent `b.lo` branches in the first encoder. After the opaque-mask fix, the retained assembly has zero conditional branches in field multiply and nine fixed-loop `b.ne` branches in the X25519 function, with no `b.lo` there. The exact pre-fix and current assembly and scanner are recorded in `ASM_AUDIT.md`. An x86-64 Linux cross-compile with Rust 1.98.0 found zero conditional jumps in field multiply and nine fixed-counter jumps in X25519. It did not run on an x86-64 host; the retained assembly and scanner are also in `ASM_AUDIT.md`.

Missing checks

The release library Cargo test waited behind the shared build lock, then was stopped after more than twenty minutes of compilation as host load reached 256; it emitted no test result. The host was heavily contended, so no wall-time comparison was taken. A Modal x86-64 validation launch was rejected by automatic approval review before source export because the new eight-file bundle lacks specific authorization. Machine-code timing behavior, isolated ARM/x86 A/A and A/B comparisons, and complete protocol regression remain open. The diagram is a flow of observed checks, not a speedup graph. The smaller source-bound Cargo harness subsequently passed 13 focused tests across the imported 25519 and P-521 modules; the full library test remains open.